The first law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a consistent sign convention in an energy balance.

SUCCESS CRITERIA - I can...

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “internal energy” and “work” to explain your model or result. Sentence starters are optional.

internal energy: energy stored in a system’s microscopic motion/interactions

work: energy transferred by a force along a displacement

surroundings: everything outside the chosen system

01 Warm-up: bring back an earlier idea

A 3 kg sample completely changes phase with latent heat 500 J/kg.
Find the energy magnitude required.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta U = Q - W_{by}$.
Q is positive into the system; W_{by} is positive when the system does work on its surroundings.

Worked reasoning: A system receives 90 J of heat and does 30 J of work on its surroundings. With ΔU=Q-W, internal energy increases by 60 J.

Your turn: Draw labeled arrows crossing a system boundary and mark the 60 J stored increase.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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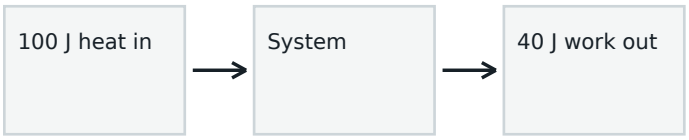
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A system absorbs 100 J of heat and does 40 J of work. **Draw** the energy arrows and find its internal-energy change.

Sentence starter: I found ... [unit]. This value means ... in this situation.



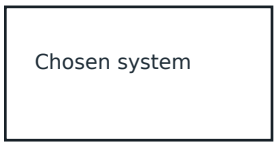
Marzano check - circle: ① ② ③

04 Apply the physics

A gas in a piston absorbs 80 J of heat and does 20 J of work by pushing outward.
Label the input, work output and stored-energy change.
Use $\Delta U = Q - W_{by}$ to find whether its internal energy increases or decreases.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why does the phrase “work is 60 J” need a direction or sign convention?

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Total kinetic energy changes from 24 J to 20 J in an isolated collision. Is it elastic? Where could the difference go?
Show the energy changes and transfers; label units.

Word help: kinetic energy = energy associated with motion

Sentence starter: Both ... . They differ in ... because ... .

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Heat input 30 J; work output 10 J.
Find ΔU.
Label energy entering, leaving or stored on the system model.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Give two different heat-input/work-output pairs that increase internal energy by 40 J.
Label energy entering, leaving or stored on the system model.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3